SEMESTER-III

UMT308: THEORY OF MACHINES				
	L	T	P	Cr
	3	1	2*	4.0

Course Objective: To introduce different types of mechanisms forming different subsystem of machines. To impart the knowledge of vector and matrix methods for position, velocity and acceleration analysis with software tools. This course deals with the dynamics of various physical systems like flywheels, governors, gyroscopes etc.

Syllabus

Kinematics of Machines: Introduction to linkages, Gears, Cam mechanics, belts and rope, as subsystems of machines.

Linkage Mechanisms: Links, kinematic pairs, degree of freedom, inversions, mechanisms, transmission angle and mechanical advantage. Graphical and analytical methods for position, velocity and acceleration analysis.

Static and Dynamic force analysis in mechanisms: Static equilibrium, Equivalent dynamical systems, Static and dynamic analysis of slider-crank mechanisms and engine force analysis.

Balancing: Balancing of rotating and reciprocating masses, single cylinder and multi-cylinder in-line engines, Field balancing of rotors.

Flywheel and Governors: Turning moment diagram of the engines, Flywheel design, Types of governors and their applications.

Gyroscopes: Gyroscopic action in automobiles, gyroscopes and their role in stabilization in ships, and airplanes.

Laboratory Work (if applicable)

Students shall perform experiments based on

1. Centrifugal force
2. Slider Crank mechanism.
3. Cam and follower mechanism.
4. Balancing of rotating and reciprocating masses
5. Gyroscopic effect

Course Learning Outcomes (CLOs)

The students will be able to:

1. select and analyze a set of mechanisms to achieve desired motion transformation.
2. apply graphical and/or analytical methods to determine position, velocity, and acceleration in planar mechanisms.
3. perform static/dynamic force analysis of mechanisms using principles of static equilibrium/equivalent dynamical systems.
4. evaluate and carry out balancing of rotors or reciprocating masses.
5. apply engineering principle of mechanics to design motion transmission devices and flywheels.
6. determine the appropriate parameters for stability of a vehicle using the concept of gyroscopic action.

Text Books

1. J. J. Uicker, G. R. Pennock, and J. E. Shigley, Theory of Machines and Mechanism, Oxford Press [2009].
2. S. S. Rattan, Theory of Machines, Tata Mc Graw Hill (2019).

Reference Books

1. Ambekar, A.G., Mechanism and Machine Theory, Prentice Hall of India, New Delhi (2011).
2. Ghosh A.K. and Mallik A.K., Theory of Mechanisms and Machines, Affiliated East-West press Pvt. Ltd. (1993).
3. Norton R.L. Kinematics & Dynamics of Machinery, McGraw Hill Education, (2017)
4. Khurmi, R.S. and Gupta J.K. Theory of Machines, S Chand (2020).
5. Grover, G.K., Mechanical Vibrations, Nem Chand and Bros, Roorkee (1996).
6. Rao, S.S., Mechanical Vibrations, Addison Wesley Publishing Company, New York (1995).

Evaluation Scheme:

Sr.No.	Evaluation Elements	Weightage (%)
1.	MST	25-30
2.	EST	40-45
3.	Sessional (May include assignments/quizzes/projects)	30